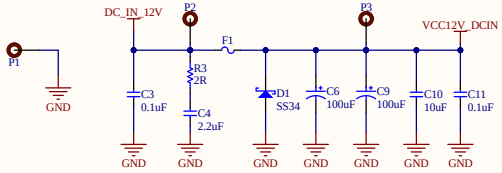
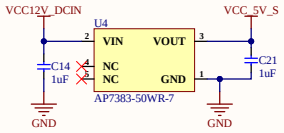
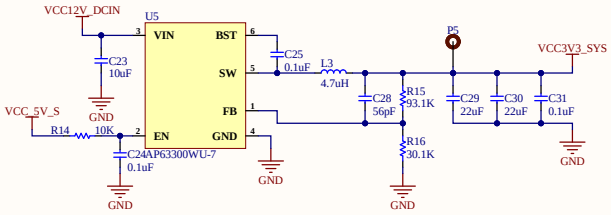
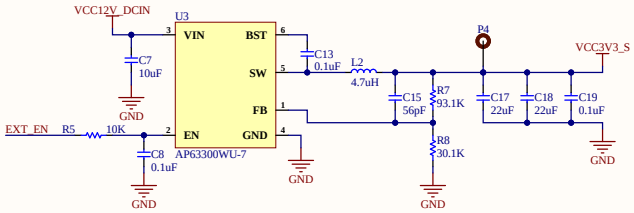
Power 3.3V_SYS



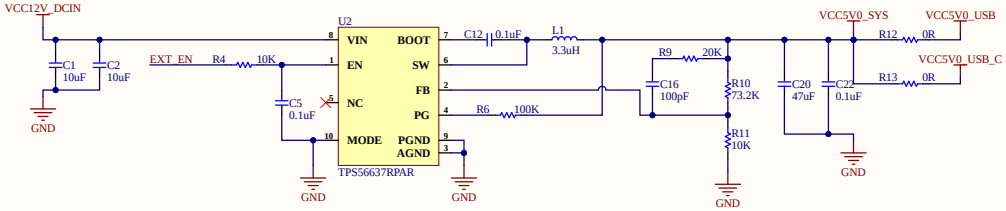


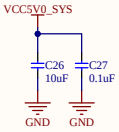


Power 3.3V_S

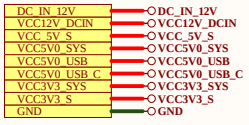


Power 5V





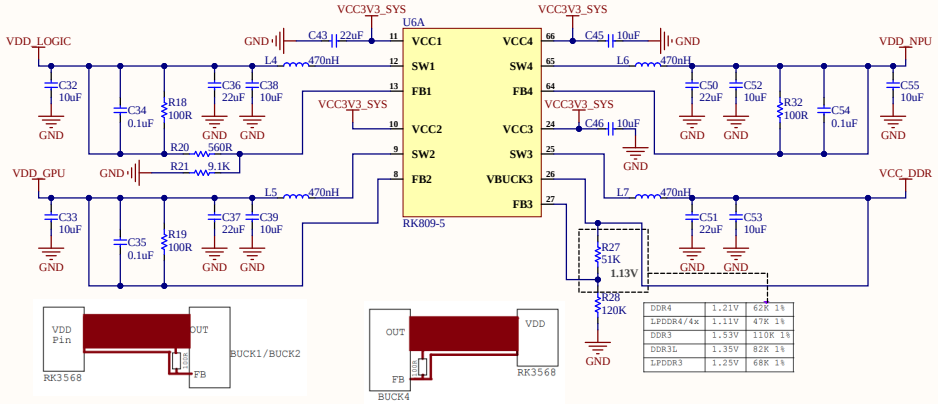




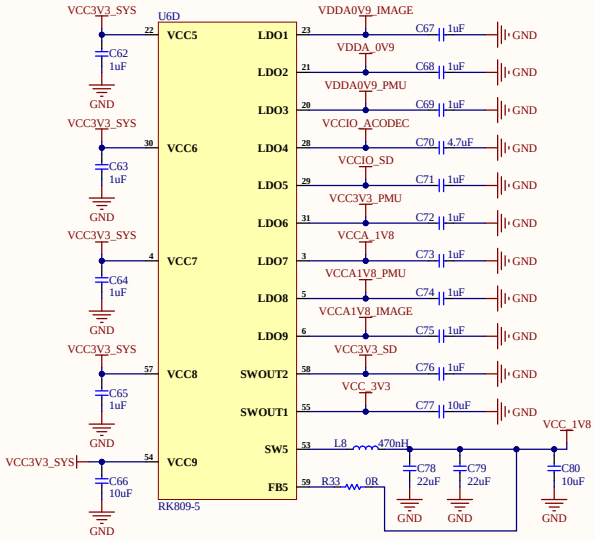
eSpark Consulting Group

Title: Power-DC-IN-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 25

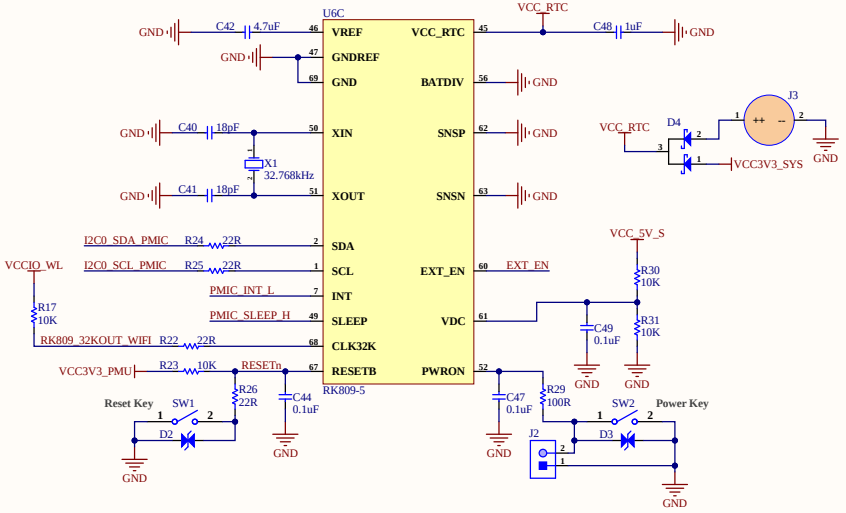
PMIC RK809 DC-DC



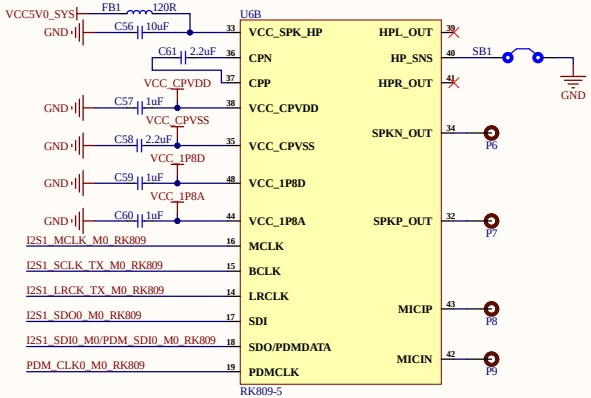
PMIC RK809 LDO



PMIC RK809 Management



PMIC RK809 CODEC



RESETn
EXT_EN
PMIC_INT_L
RK809_32KOUT_WIFI
PMIC_SLEEP_H

I2C0
I2C0_SDA_PMIC
I2C0_SCL_PMIC

I2S1
I2S1_MCLK_M0_RK809
I2S1_SCLK_TX_M0_RK809
I2S1_LRCK_TX_M0_RK809
I2S1_SDO0_M0_RK809
I2S1_SDI0_M0/PDM_SDI0_M0_RK809
PDM_CLK0_M0_RK809

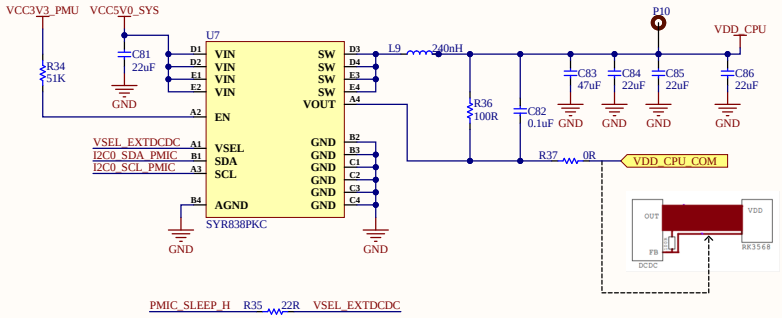
VCC3V3_SYS
VDD_LOGIC
VDD_GPU
VDD_NPU
VCC_DDR
VDDA0V9_IMAGE
VDDA_0V9
VDDA0V9_PMU
VCCIO_ACODEC
VCCIO_SD
VCC_5V_S

VCC3V3_PMU
VCCA_1V8
VCCA1V8_PMU
VCCA1V8_IMAGE
VCC3V3_SD
VCC_3V3
VCC_1V8
VCC5V0_SYS
VCCIO_WL
GND

eSpark Consulting Group

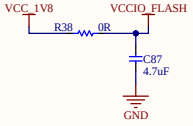
Title: Power-PMIC-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 25

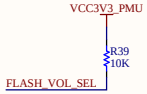
VDD_CPU



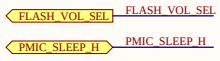
	VCCIO2 domain voltage: Recommend voltage value (VCCIO_FLASH)	FLASH_VOL_SEL state decided to VCCIO2 domain IO driven by default
eMMC	1.8V	FLASH_VOL_SEL --> Logic=H
Nand flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)
SPI flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)

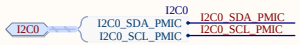
Flash Power Manage

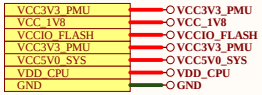




Note:
FLASH_VOL_SEL state decided
to VCCIO2 domain IO driven by default
Logic=L:3.3V IO driven
Logic=H:1.8V IO driven



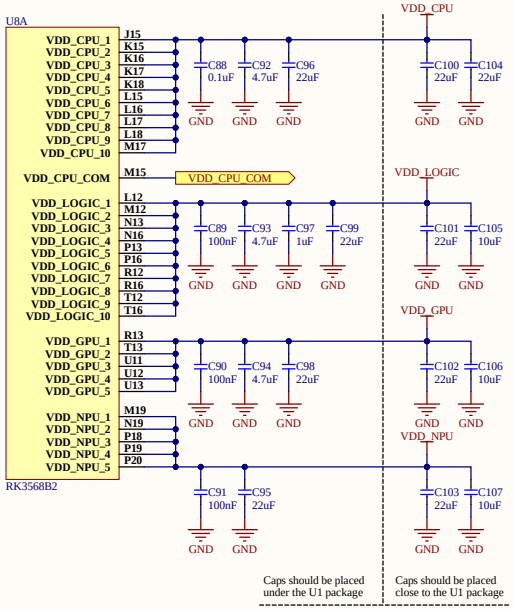


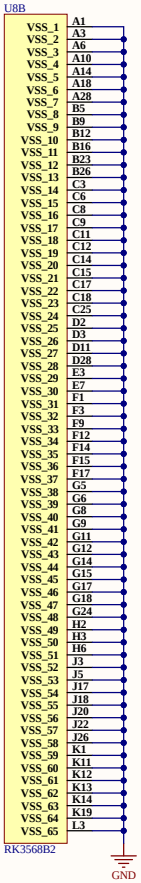


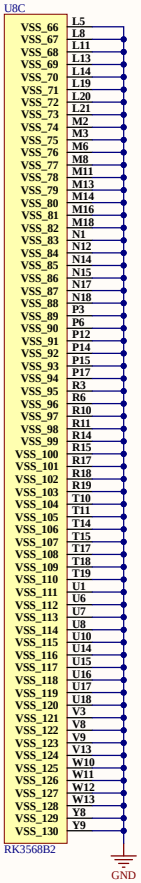
eSpark Consulting Group

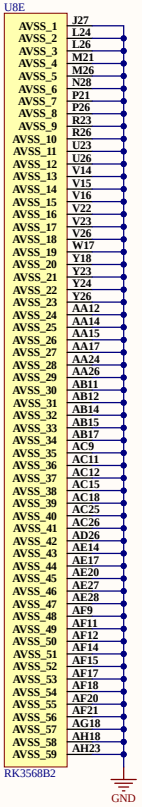
Title: Power-Others-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 25

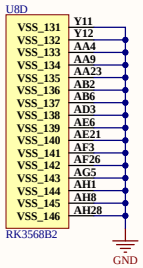
RK3568 Power & GND

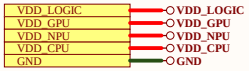








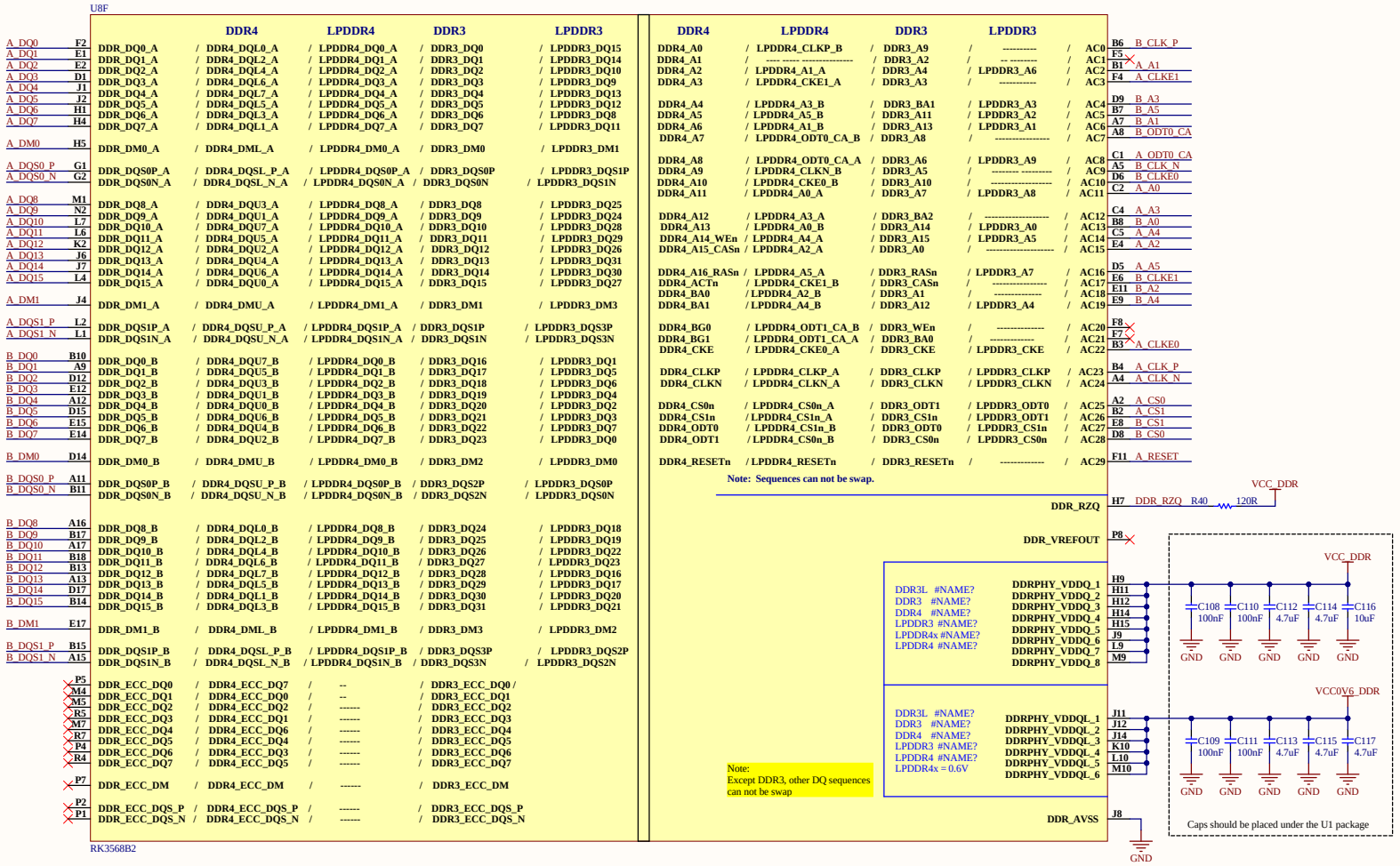


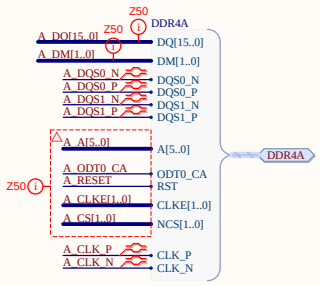


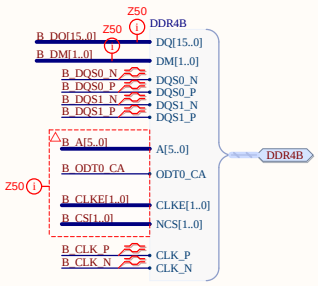
eSpark Consulting Group

Title: RK3568-Power-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 25

RK3568 DDR PHY

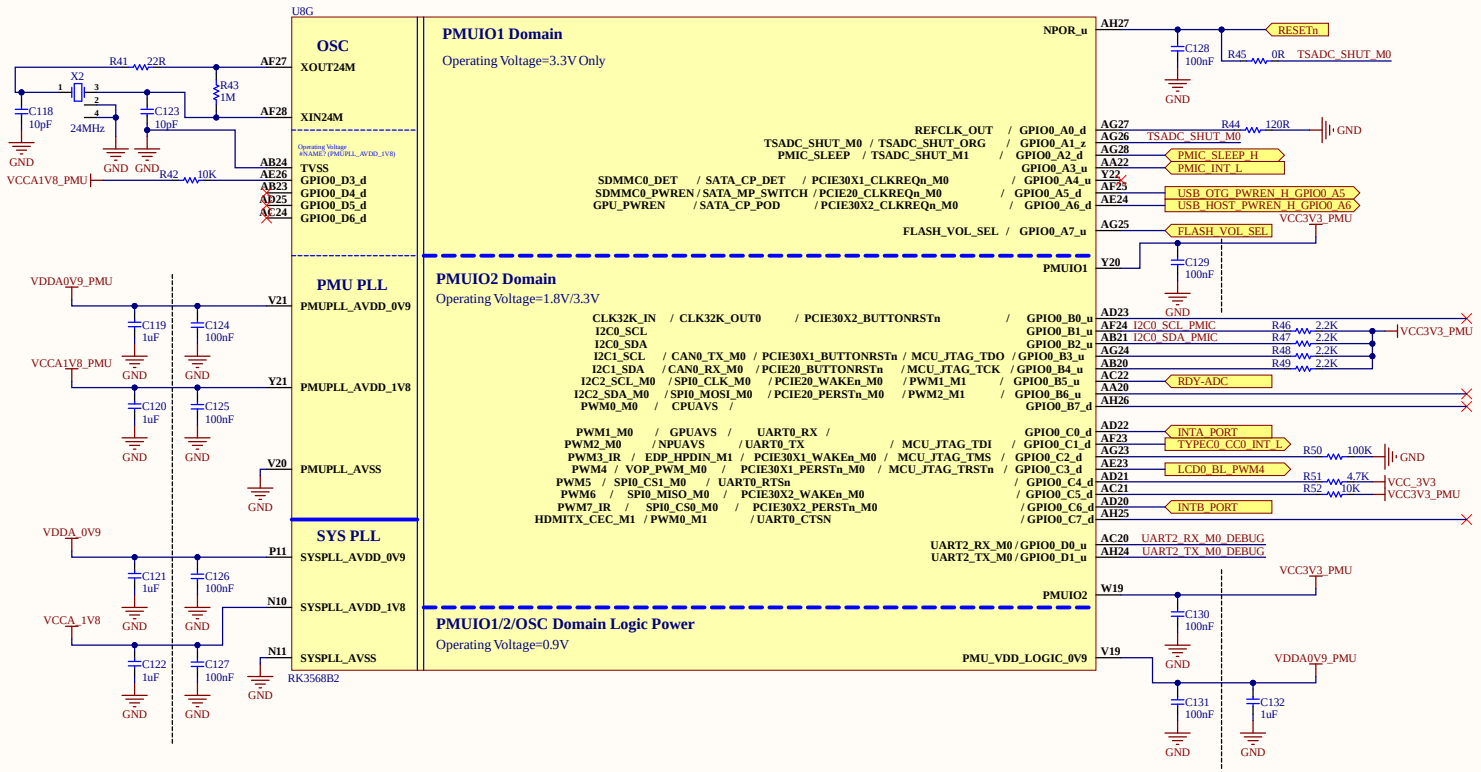






Title:	RK3568-DDR-PHY.SchDoc
File:	MVP-Dash-v1.0.PrjPcb
Date: 7/18/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 25

RK3568 OSC/PLL/PMUIO1/2



Note:
Caps of between dashed black lines and U1 should be placed under the U1 package. Other caps should be placed close to the U1 package.

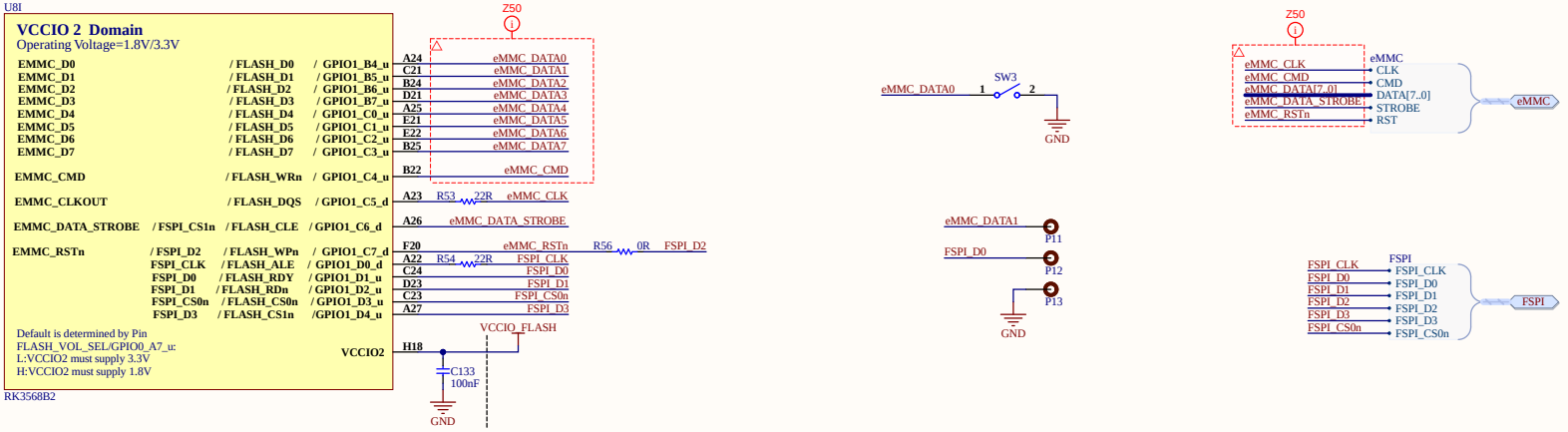
UART2_DEBUG ↔ UART2_DEBUG, UART2_RX_M0_DEBUG, UART2_TX_M0_DEBUG

I2C0 ↔ I2C0_SDA_PMIC, I2C0_SCL_PMIC

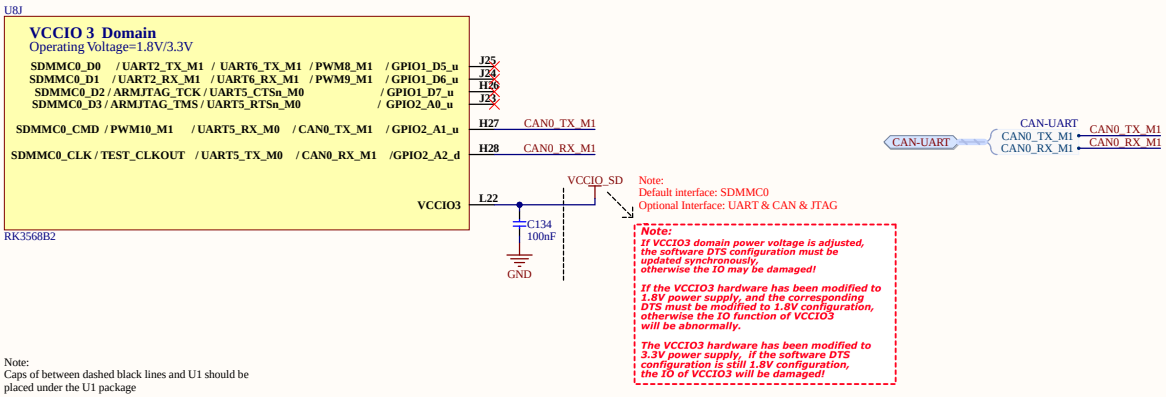
VCCA1V8_PMU, VDDA_0V9, VCCA_1V8, VCC3V3_PMU, VDDA0V9_PMU, GND

Title:	RK3568-OSC-PMUIO-Sheet.SchDoc
File:	MVP-Dash-v1.0.PrjPcb
Date: 7/18/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 25

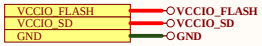
RK3568 Flash (VCCIO2 Domain)



RK3568 SD (VCCIO3 Domain)



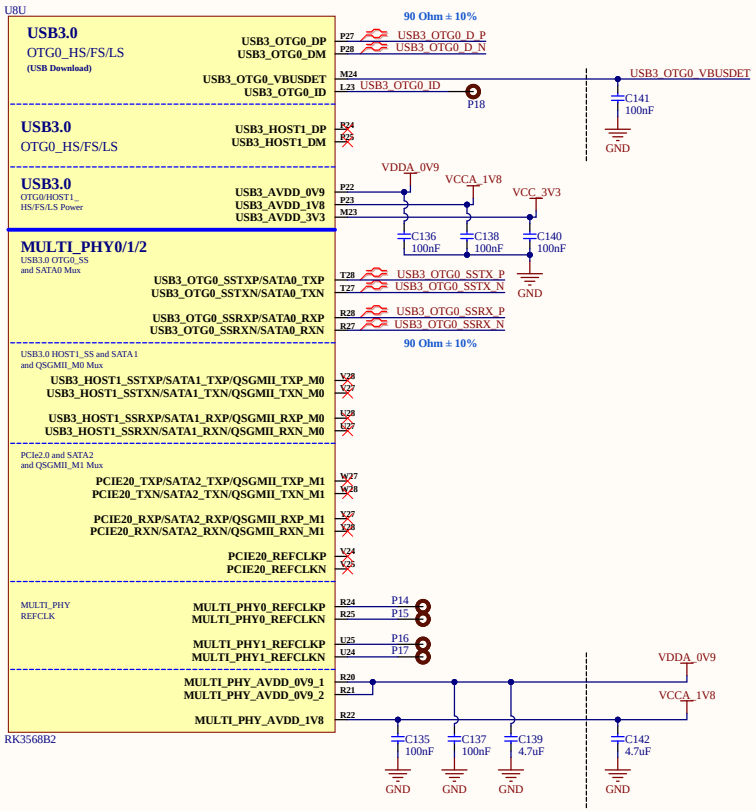
Note:
Caps of between dashed black lines and U1 should be placed under the U1 package



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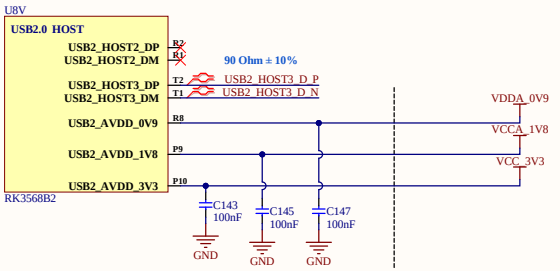
Title: RK3568-Flash-SD-Controller.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 25

RK3568 USB3.0/SATA/QSGMII/PCIe2.0 x1

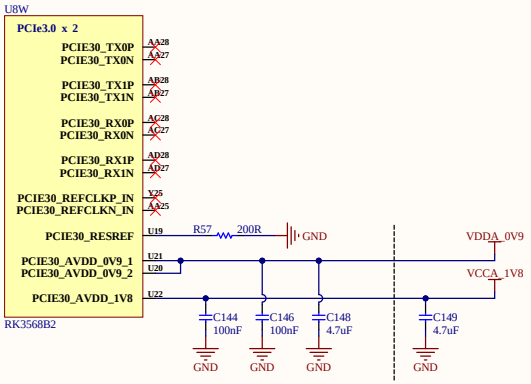


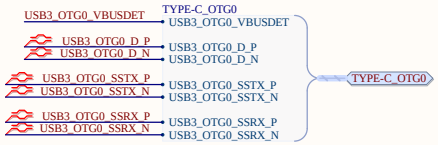
Note:
Caps of between dashed black lines and U1 should be placed under the U1 package. Other caps should be placed close to the U1 package.

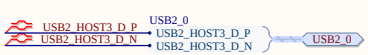
RK3568 USB2.0 HOST

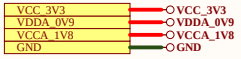


RK3568 PCIe3.0 x2





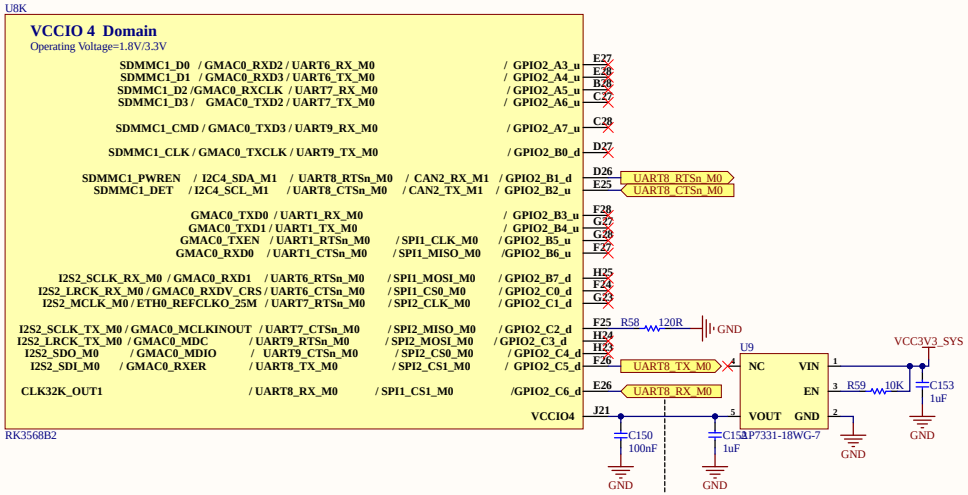




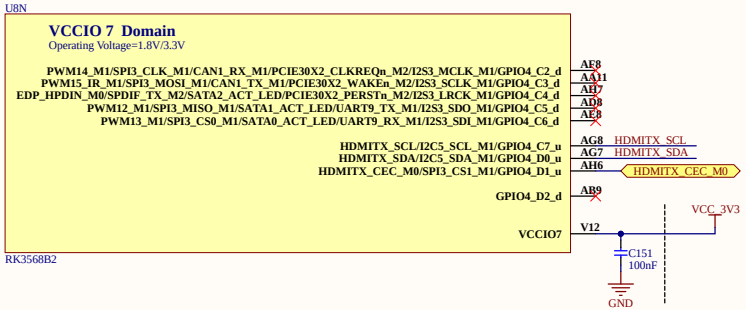
eSpark Consulting Group

Title: RK3568-USB-PCIe-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 25

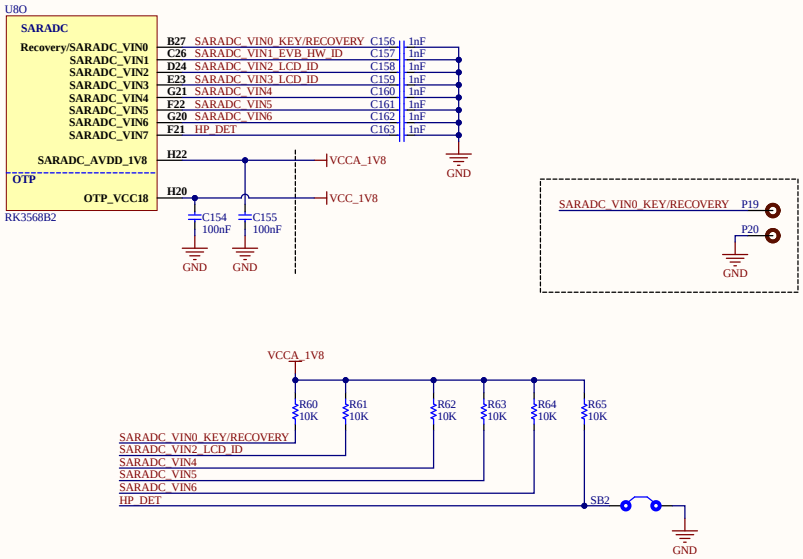
RK3568 (VCCIO4 Domain)



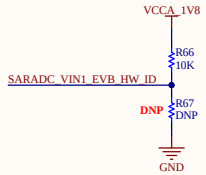
RK3568 (VCCIO7 Domain)



RK3568 (SARADC/OTP)

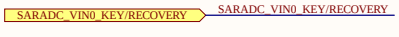


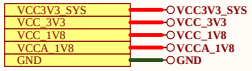
SARADC_VIN1_EVB_HW_ID	Rup	Rdown	ADC	
EVB1	10K	DNP	1023	1.8V
EVB2	20K	100K	852	1.5V
EVB3	18K	36K	681	1.2V
EVB4	51K	51K	512	0.9V
EVB5	36K	18K	340	0.6V
EVB6	100K	20K	170	0.3V
EVB7	DNP	10K	0	0V
EVB8				



Note:
Caps of between dashed black lines and U1 should be placed under the U1 package.



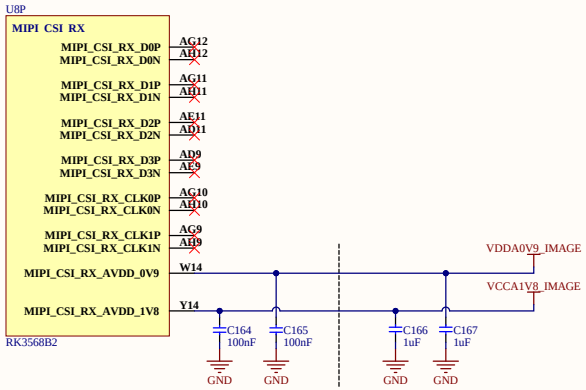




eSpark Consulting Group

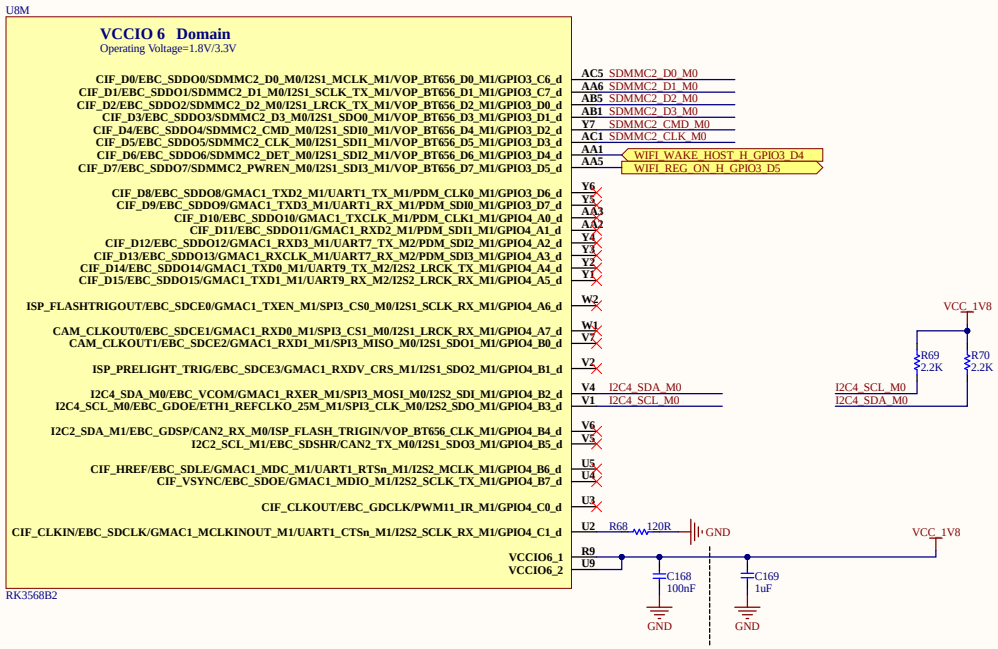
Title: RK3568-SARADC-GPIO.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 25

RK3568 (MIPI CSI RX)



Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

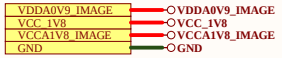
RK3568 (VCCIO6 Domain)



Mode	16bit	12bit	10bit	8bit
CIF_D0	D0	--	--	--
CIF_D1	D1	--	--	--
CIF_D2	D2	--	--	--
CIF_D3	D3	--	--	--
CIF_D4	D4	D0	--	--
CIF_D5	D5	D1	--	--
CIF_D6	D6	D2	D0	--
CIF_D7	D7	D3	D1	--
CIF_D8	D8	D4	D2	D0
CIF_D9	D9	D5	D3	D1
CIF_D10	D10	D6	D4	D2
CIF_D11	D11	D7	D5	D3
CIF_D12	D12	D8	D6	D4
CIF_D13	D13	D9	D7	D5
CIF_D14	D14	D10	D8	D6
CIF_D15	D15	D11	D9	D7

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

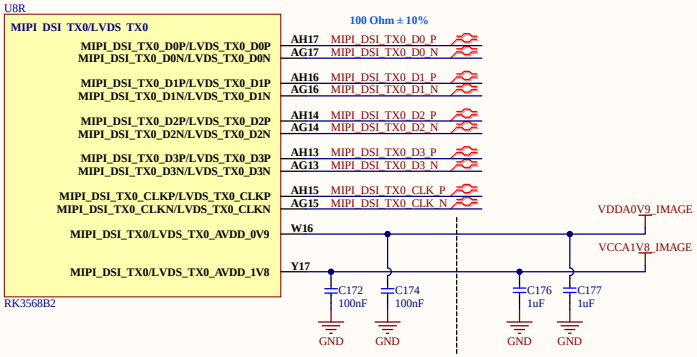




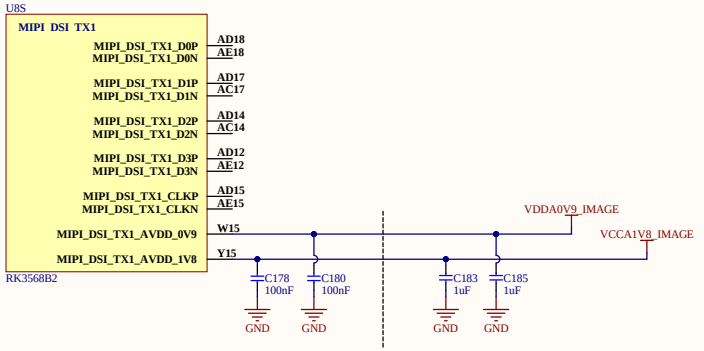
eSpark Consulting Group

Title: RK3568-VI-Interface-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 25

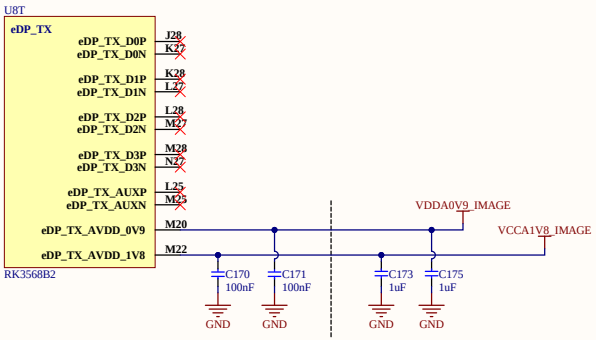
RK3568 (MIPI DSI TX0)



RK3568 (MIPI DSI TX1)

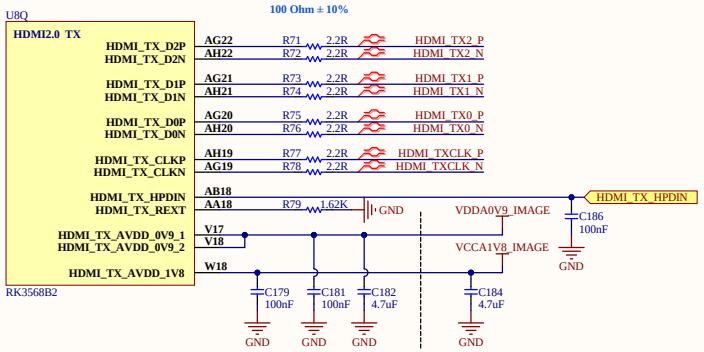


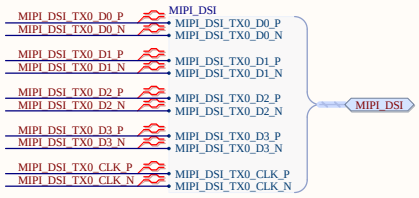
RK3568 (eDP TX)

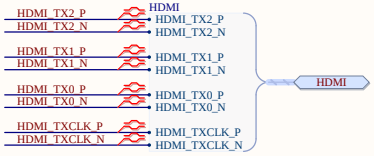


Note:
Caps of between dashed black lines and U1 should be placed under the U1 package. Other caps should be placed close to the U1 package.

RK3568 (HDMI2.0 TX)





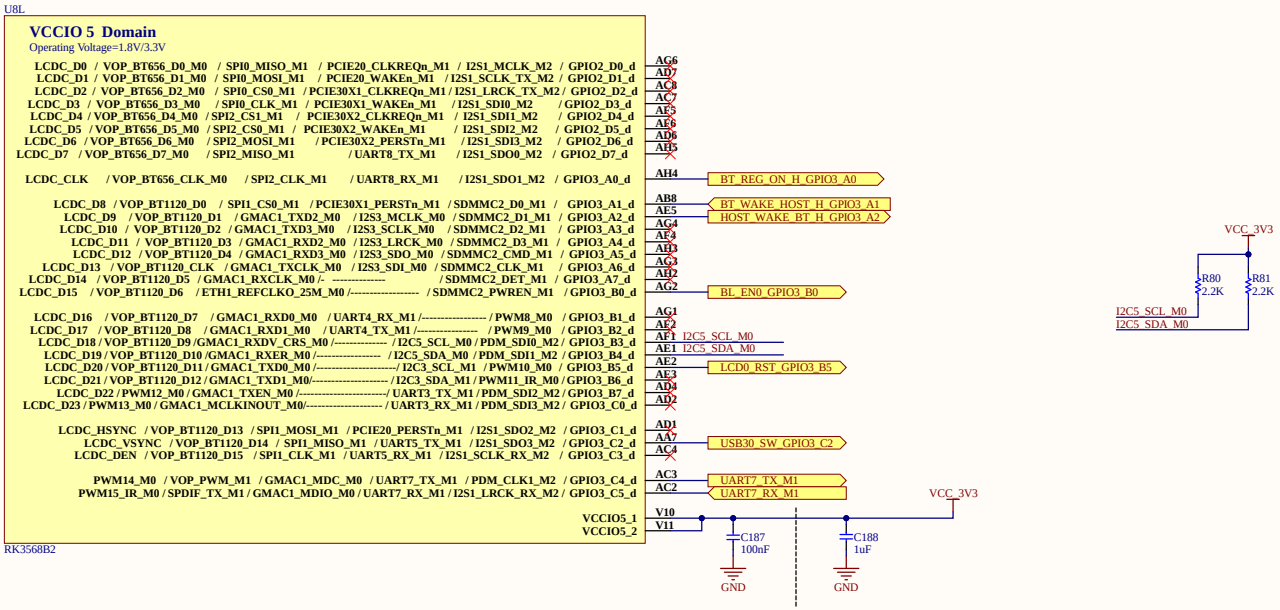




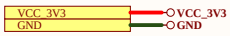
eSpark Consulting Group

Title: RK3568-VO-Interface-1.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 25

RK3568 (VCCIO5 Domain)

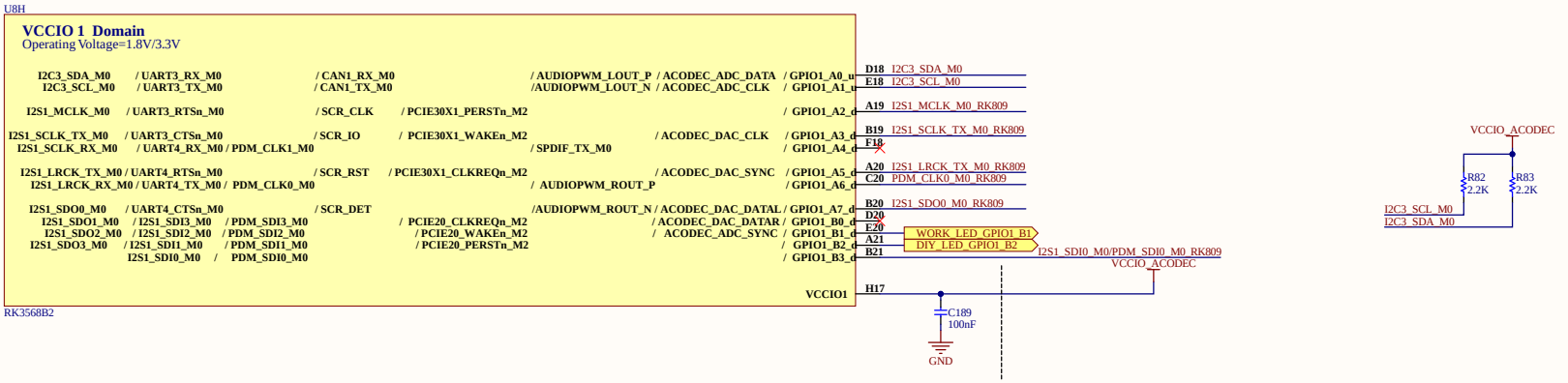


Note:
Caps of between dashed black line and U1 should be placed under the U1 package

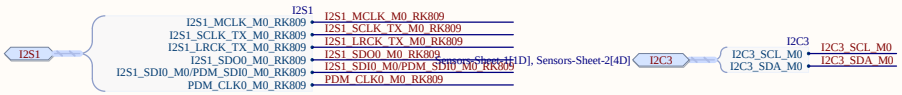


Title: RK3568-VO-Interface-2.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 25

RK3568 (VCCIO1 Domain)



Note:
Caps of between dashed black line and U1 should be placed under the U1 package

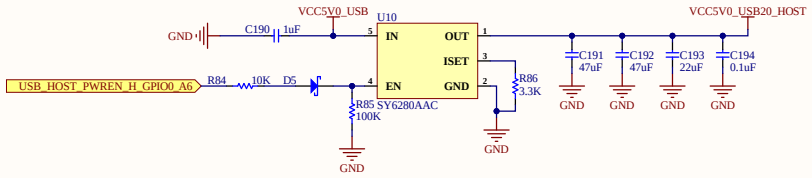


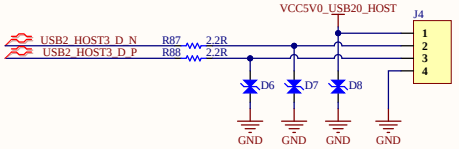


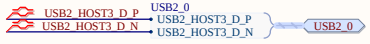
eSpark Consulting Group

Title: RK3568-Audio-Interface.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 13
Drawn By: Huzaifa Ali	Page	Total: 25

USB2.0 HOST 3





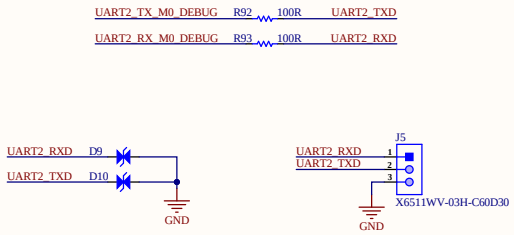




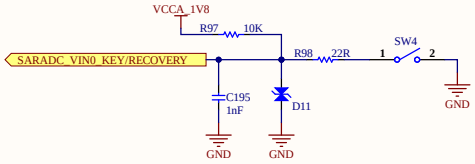
eSpark Consulting Group

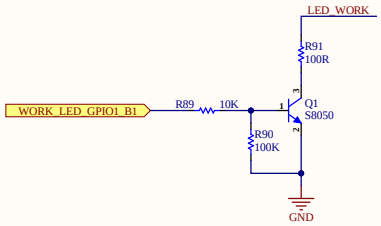
Title: USB-Ports.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/31/2024	Page	Num: 14
Drawn By: Huzaifa Ali	Page	Total: 25

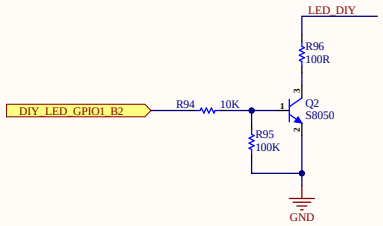
Debug

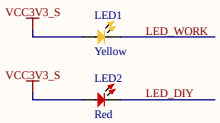


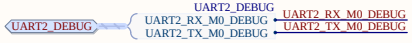
Recovery Key









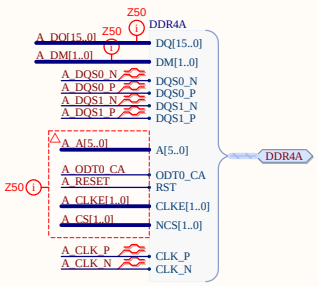


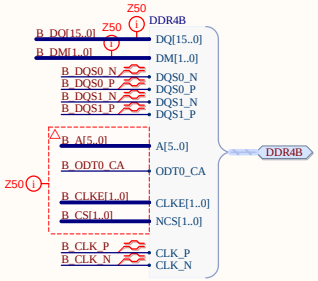


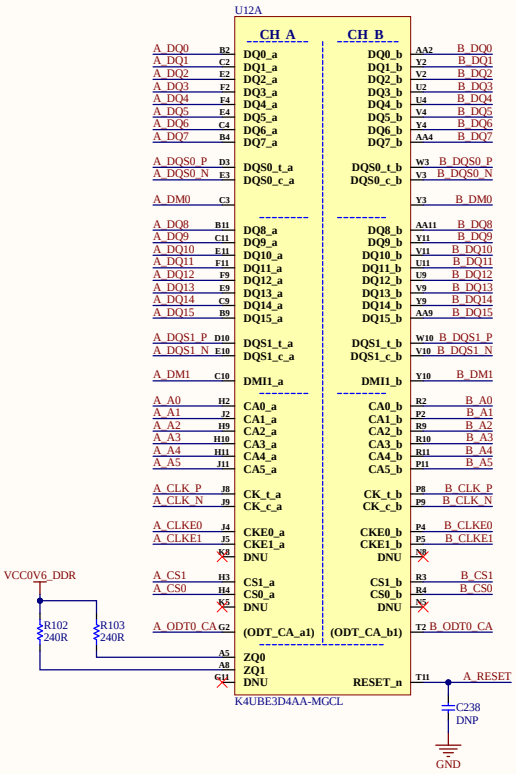
eSpark Consulting Group

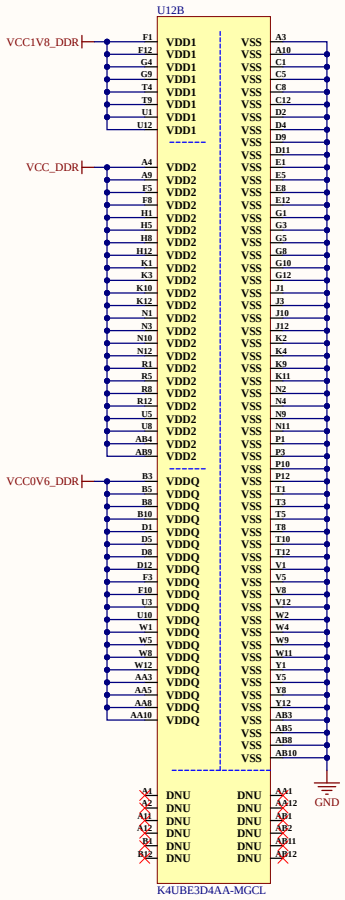
Title: LED-Debug-Sheet.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 15
Drawn By: Huzaifa Ali	Page	Total: 25

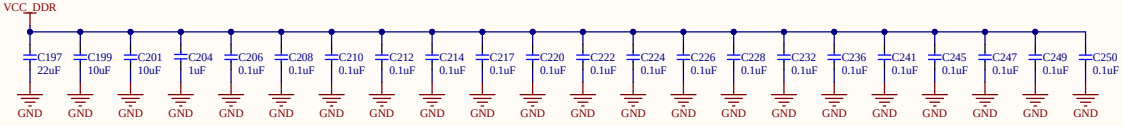
LPDDR4X

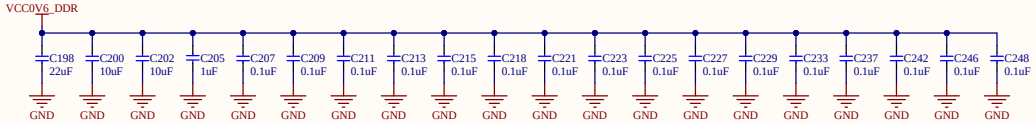


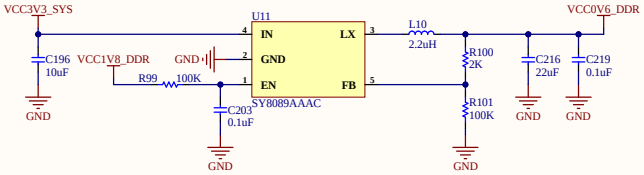


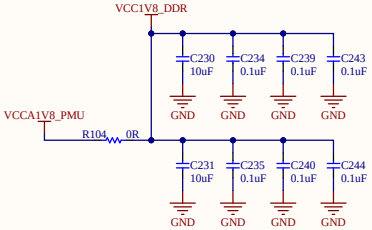


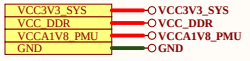








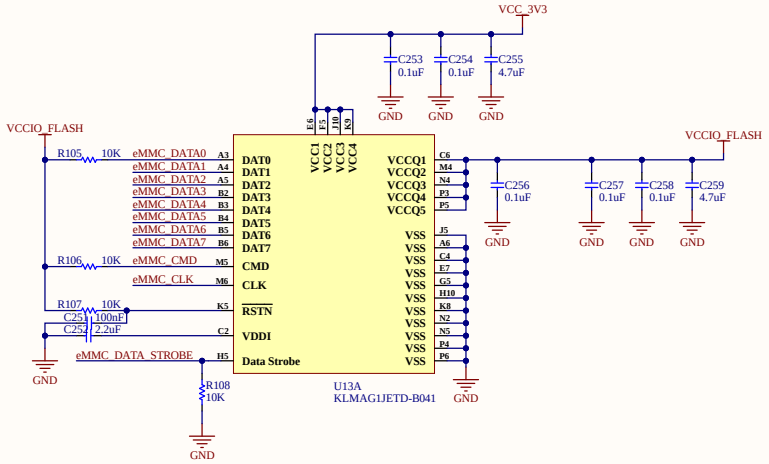


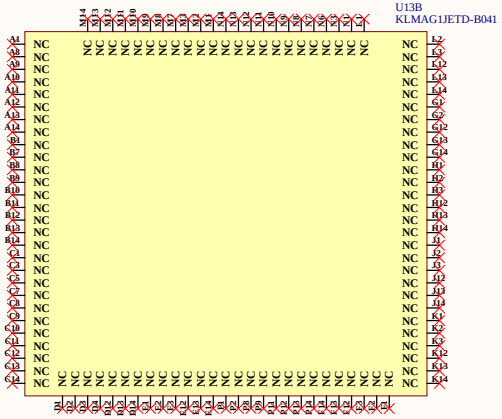


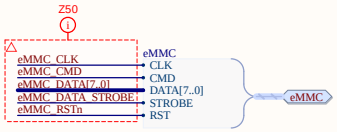
eSpark Consulting Group

Title: DRAM-LPDDR4x-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 16
Drawn By: Huzaifa Ali	Page Total: 25

eMMC





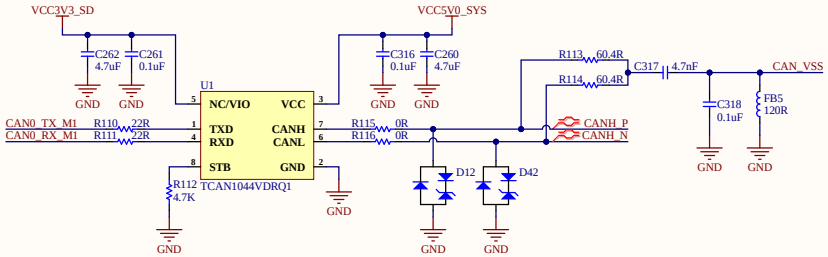




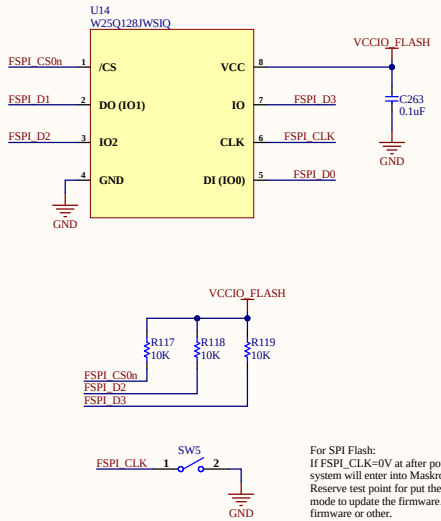
eSpark Consulting Group

Title: eMMC-Sheet.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 17
Drawn By: Huzaifa Ali	Page	Total: 25

CAN Bus



SPI Flash



For SPI Flash:
If FSPI_CLK=0V at after power on and reset, then system will enter into Maskrom mode.
Reserve test point for put the system into Maskrom mode to update the firmware. When writing mismatched firmware or other.

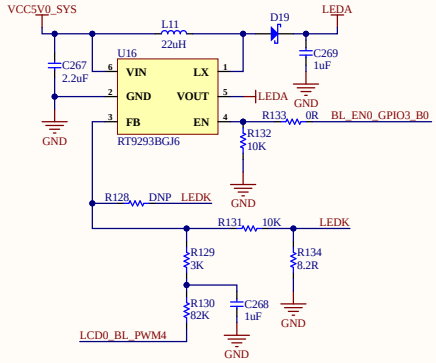


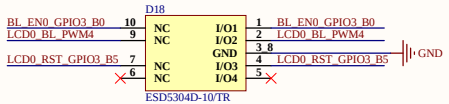
Title: CAN-Bus-SPI-Flash-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/18/2024	Page Num: 18
Drawn By: Huzaifa Ali	Page Total: 25

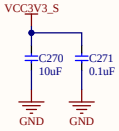
MIPI Display

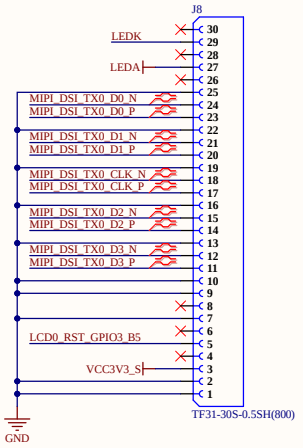
KD050FHFID103

NO.	Pin Name
1	NC
2	LEDK
3	NC
4	LEDA
5	NC
6	GND
7	MIPI_D0N
8	MIPI_D0P
9	GND
10	MIPI_D1N
11	MIPI_D1P
12	GND
13	MIPI_CLN
14	MIPI_CLP
15	GND
16	MIPI_D2N
17	MIPI_D2P
18	GND
19	MIPI_D3N
20	MIPI_D3P
21	GND
22	GND
23	BIST
24	GND
25	NC
26	RESET
27	NC
28	VCI
29	GND
30	GND











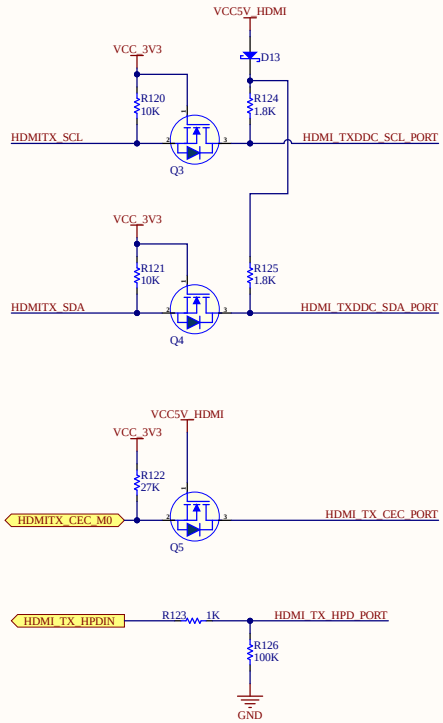


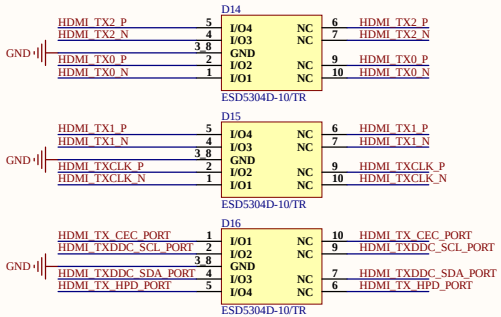


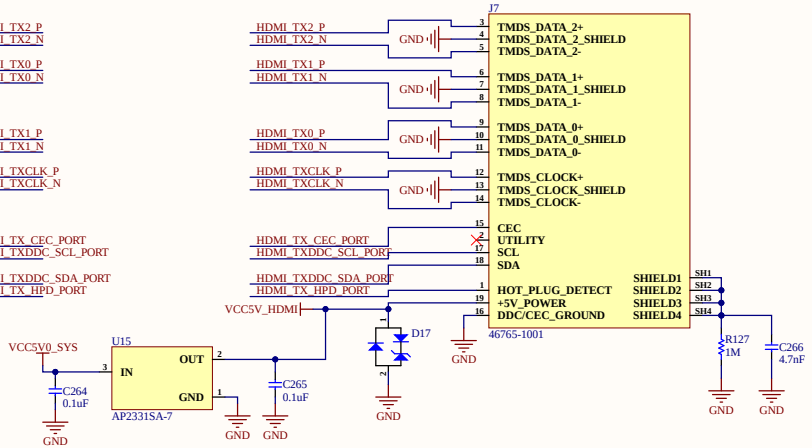
eSpark Consulting Group

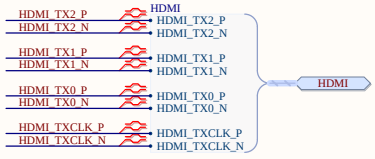
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File: MVP-Dash-v1.0.PrjPcb	
Date: 7/19/2024	Page Num: 19
Drawn By: Huzaifa Ali	Page Total: 25

HDMI 2.0









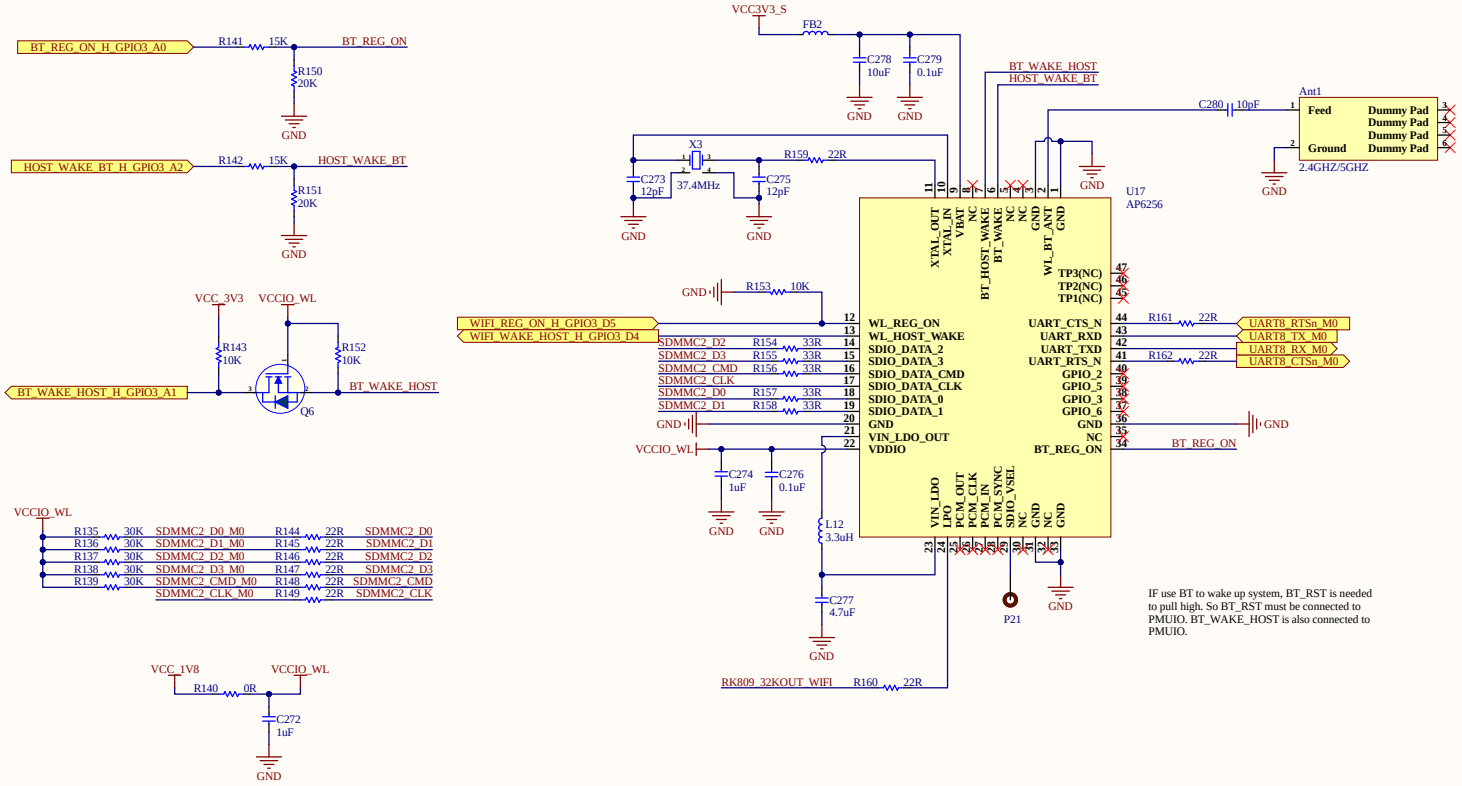




eSpark Consulting Group

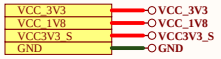
Title: VO-HDMI2.0-TX-Sheet.SchDoc	
File: MVP-Dash-v1.0.PrjPcb	
Date: 7/31/2024	Page Num: 20
Drawn By: Huzaifa Ali	Page Total: 25

Dual Band WiFi



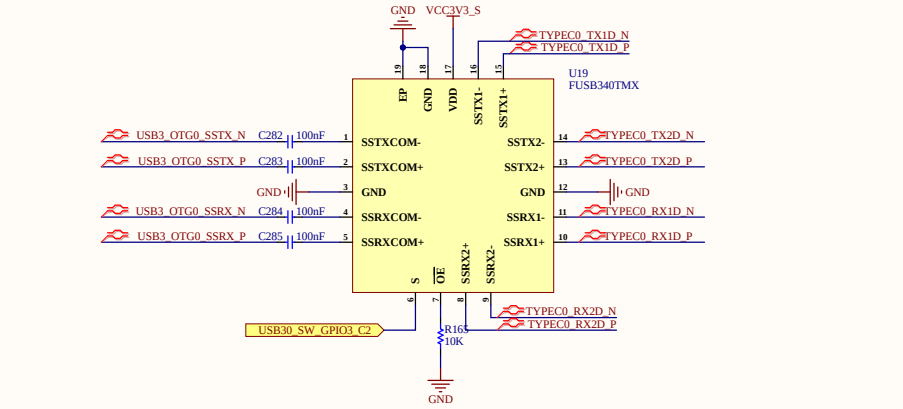


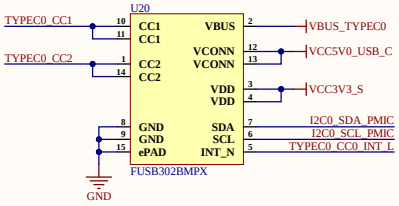


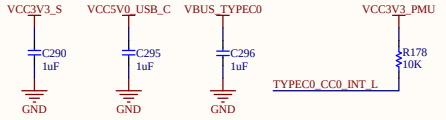


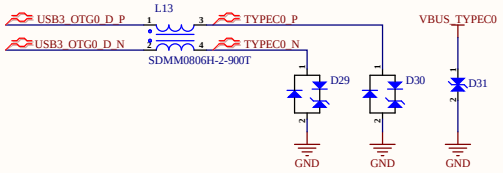
eSpark Consulting Group

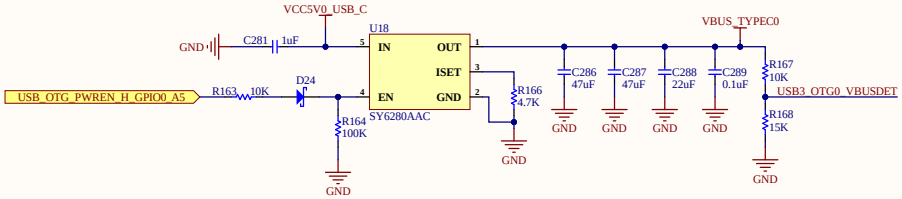
Title: WIFI-BT-SDIO-Sheet.SchDoc		
File: MVP-Dash-v1.0.PrjPcb		
Date: 7/18/2024	Page	Num: 21
Drawn By: Huzaifa Ali	Page	Total: 25

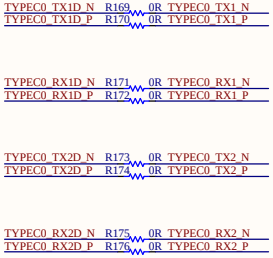




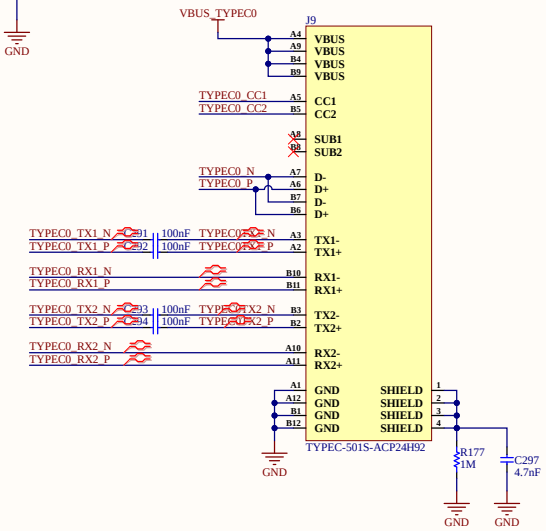




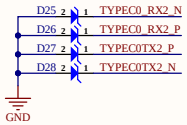


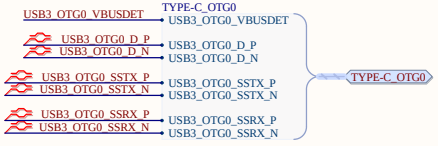


Programming Port



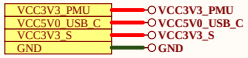








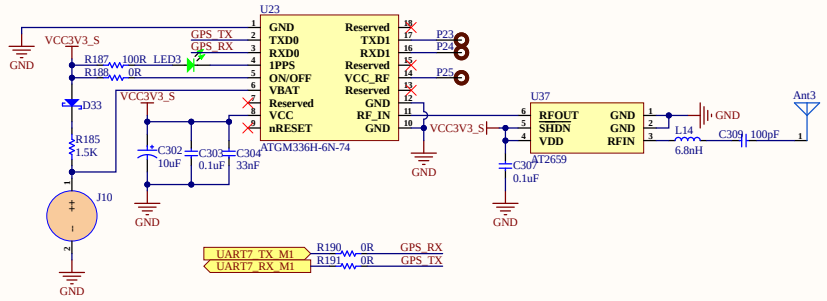




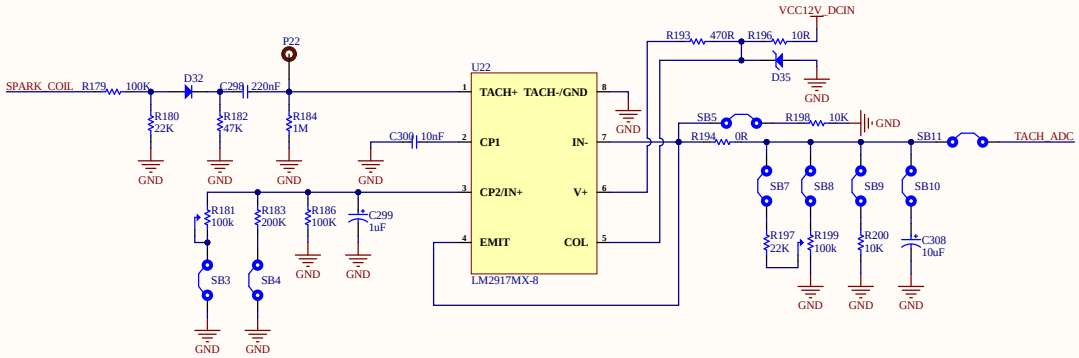
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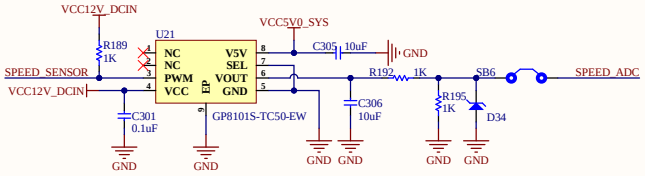
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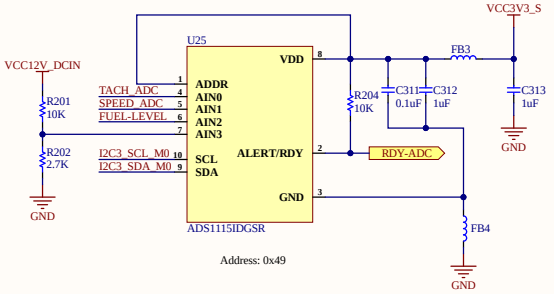
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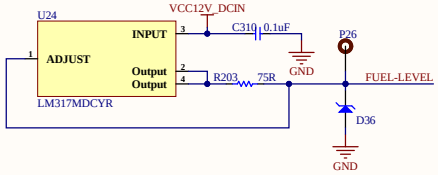
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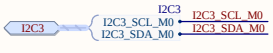


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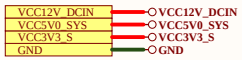


FUEL LEVEL SENSOR





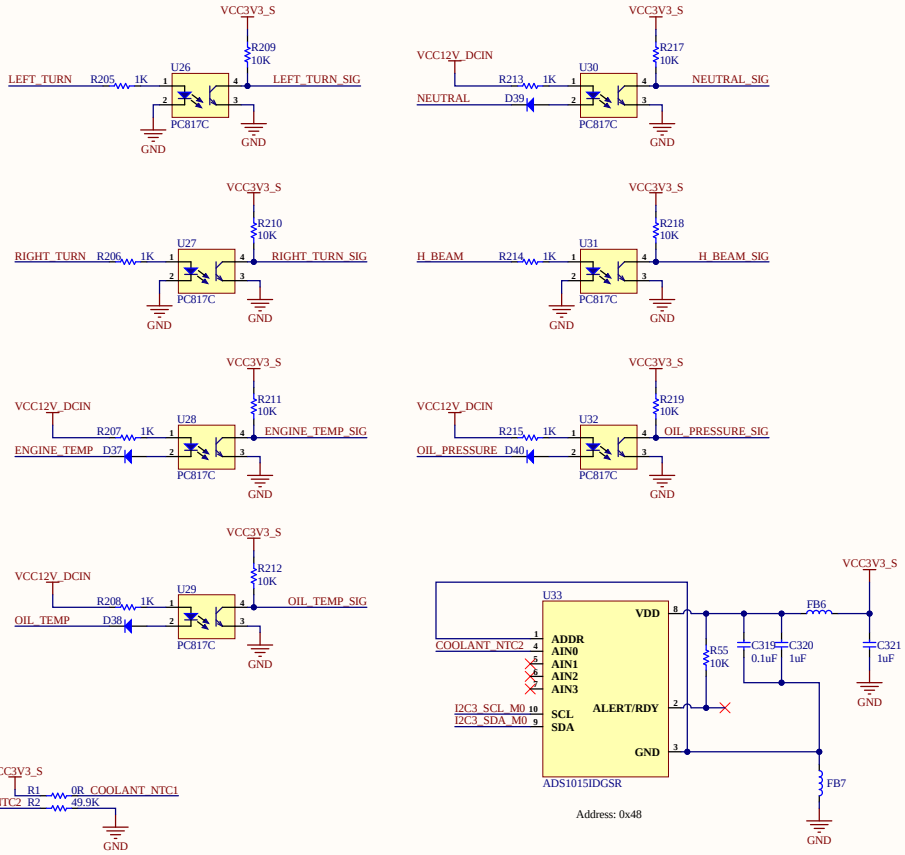




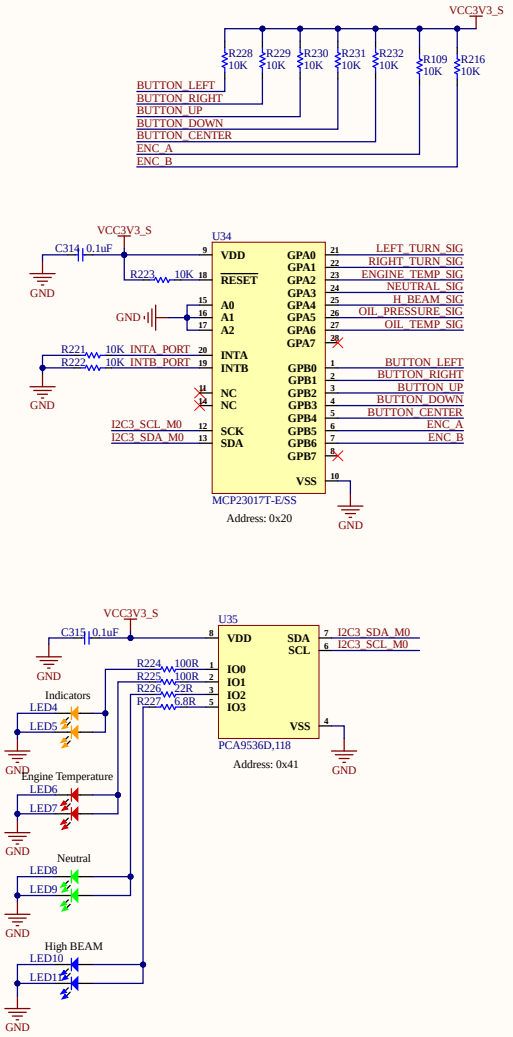
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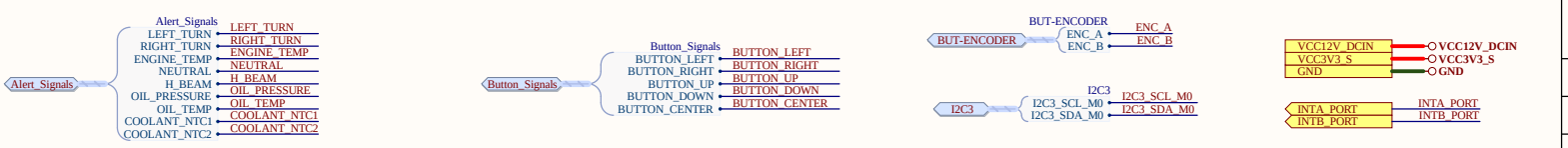
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Alert Signals



Buttons Signals

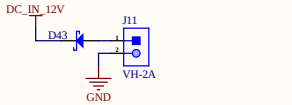


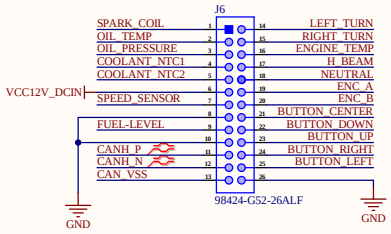


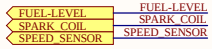
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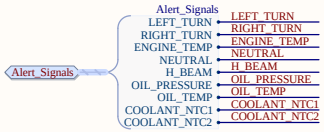
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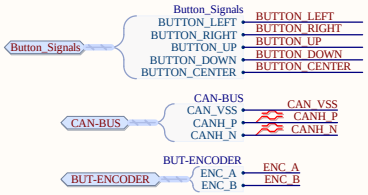
DC Input













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